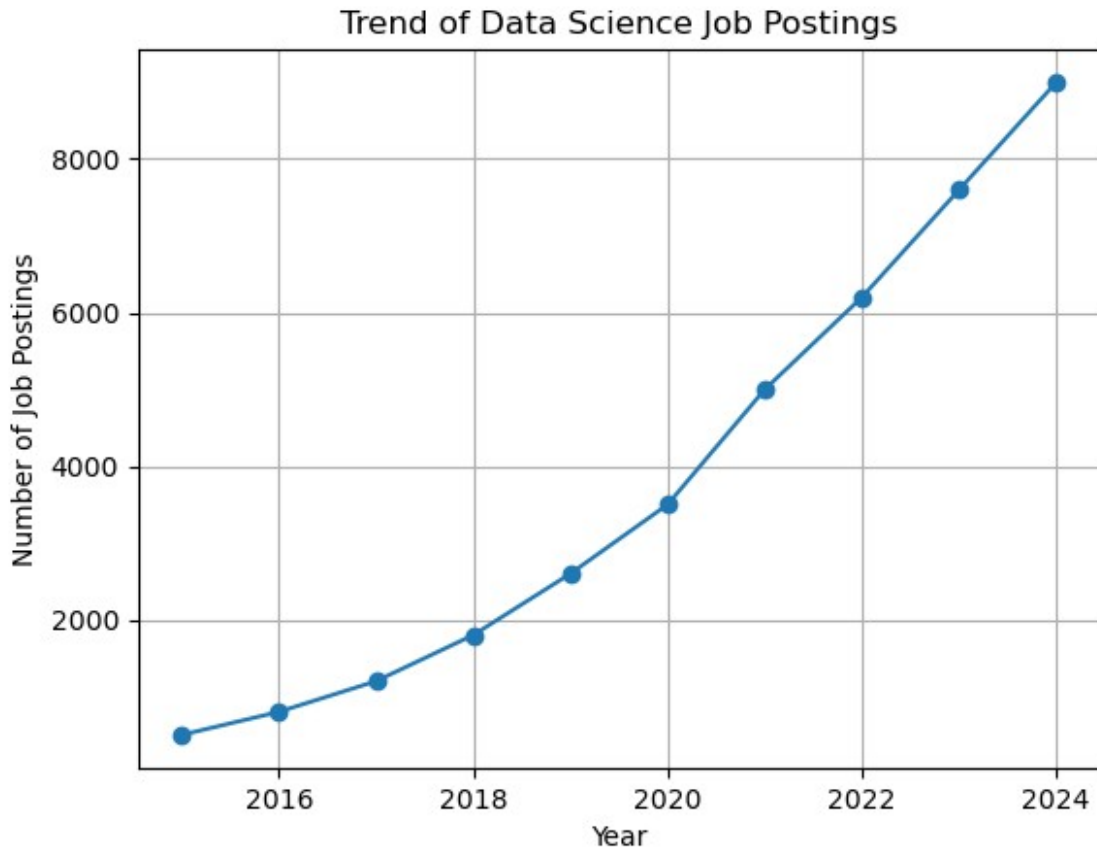


Name:Swathi Sri R Register Number: 250701769 Experiment Name: Exploratory Data Analysis to Develop Deep Insights from a Dataset

1.a. Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Year": [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023,
2024],
    "Job_Postings": [500, 800, 1200, 1800, 2600, 3500, 5000, 6200,
7600, 9000]
}
df=pd.DataFrame(data)
print(df)
plt.plot(df["Year"], df["Job_Postings"], marker='o')
plt.title("Trend of Data Science Job Postings")
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.grid(True)
plt.show()
```

	Year	Job_Postings
0	2015	500
1	2016	800
2	2017	1200
3	2018	1800
4	2019	2600
5	2020	3500
6	2021	5000
7	2022	6200
8	2023	7600
9	2024	9000

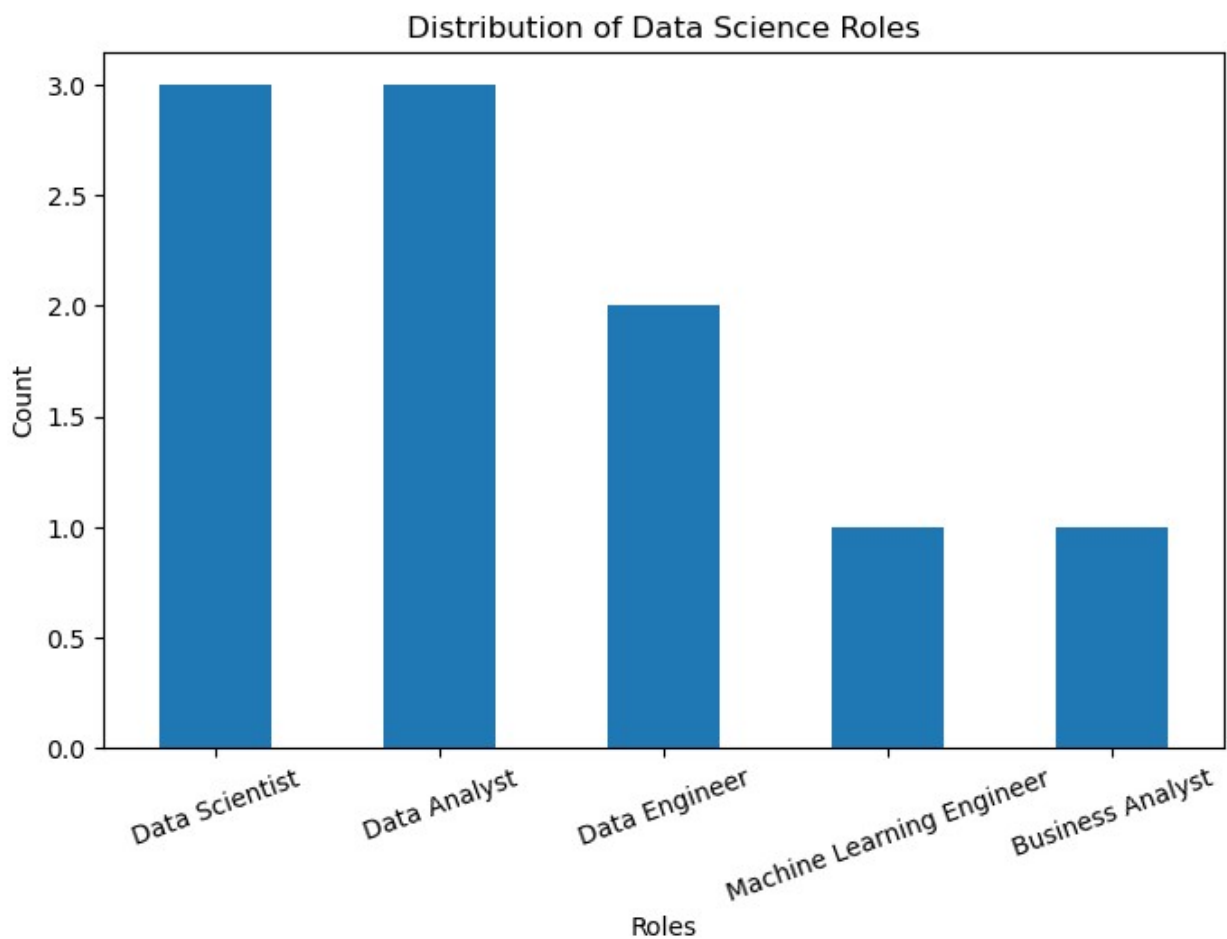


1.b. Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Role": [
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Machine Learning Engineer",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Business Analyst",
        "Data Analyst"
    ]
}
df = pd.DataFrame(data)
df
role_count = df["Role"].value_counts()
print(role_count)
```

```
plt.figure(figsize=(8,5))
role_count.plot(kind="bar")
plt.title("Distribution of Data Science Roles")
plt.xlabel("Roles")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.show()
```

```
Role
Data Scientist      3
Data Analyst        3
Data Engineer       2
Machine Learning Engineer  1
Business Analyst    1
Name: count, dtype: int64
```



1.c. Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

```

import pandas as pd
structured_data = {
    "ID": [101, 102, 103],
    "Name": ["Ravi", "Priya", "Arun"],
    "Age": [20, 21, 22]
}
df = pd.DataFrame (structured_data)
print("Structured Data:")
print(df)
print()
semi_structured_data = {
    "Student": {
        "ID": 101,
        "Name": "Ravi",
        "Department": "CSE"
    }
}
print("Semi-Structured Data:")
print(semi_structured_data)
print()
unstructured_data = """
Artificial Intelligence is transforming industries.
Images, videos, audio files, and text documents are examples of
unstructured data.
"""
print("Unstructured Data:")
print(unstructured_data)

```

Structured Data:

	ID	Name	Age
0	101	Ravi	20
1	102	Priya	21
2	103	Arun	22

Semi-Structured Data:

```
{'Student': {'ID': 101, 'Name': 'Ravi', 'Department': 'CSE'}}
```

Unstructured Data:

Artificial Intelligence is transforming industries.
 Images, videos, audio files, and text documents are examples of
 unstructured data.

1.d. Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

```

from cryptography.fernet import Fernet
key=Fernet.generate_key()
print("Secret Key:", key)

```

```
cipher=Fernet(key)
message = "Password123"
encrypted_message=cipher.encrypt(message.encode())
print("\nEncrypted Message:")
print(encrypted_message)
decrypted_message = cipher.decrypt(encrypted_message)
print("\nDecrypted Message:")
print(decrypted_message.decode())
```

Secret Key: b'RwIN0HBJ7m0Fh0DD-lVVkW7Wcs3ykQbd0u6VChr40g0='

Encrypted Message:

b'gAAAAABqb1-
gvxq7IUyo5AZcTNVy6zt5aDaLU0dmjWBWFaxq6D70JLC6qIcN8hdu5TAIIILfd62gbBjyl
wkhEpKxIpQQQJpPka=='

Decrypted Message:

Password123